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Synopsis

SMART SMPS MONITORING SYSTEM

Submitted in partial fulfilment of the requirements

of the degree of

Bachelor of Engineering
In
Electronics and Computer Science Engineering

by

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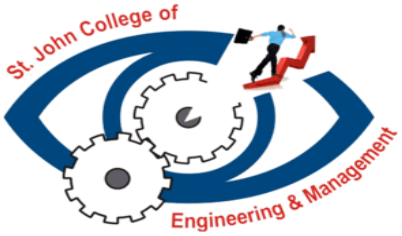
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DEPARTMENT OF ELECTRONICS AND COMPUTER SCIENCE ENGINEERING
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We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. we also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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ABSTRACT

SMART SMPS MONITORING SYSTEM is an industry-oriented embedded electronic instrument proposed for monitoring and basic testing of DC outputs from switched-mode power supplies (SMPS). The system is designed around an ESP32 controller, an INA226 high-side voltage/current/power monitoring interface, a 3.2-inch ILI9341 SPI TFT display, status LEDs and physical push buttons. The primary objective is to provide a compact and dedicated instrument that can measure output voltage and current, calculate output power using $P = V \times I$, display the measurements locally and classify the observed output into normal, warning or fault states according to a selected SMPS profile.

The project is being developed with practical exposure to an SMPS manufacturing environment. Typical target profiles include supplies such as 5 V/2 A, 12 V/5 A and 24 V/10 A; however, the actual current prototype is not claimed to support these ratings universally. In particular, the INA226 device used in the current architecture has a 36 V bus-voltage limit, so direct 60 V operation is excluded from the present prototype. Higher-voltage and higher-current operation will require a redesigned sensing, protection, switching and thermal architecture.

The present hardware integration includes the ESP32, INA226 I2C interface, TFT SPI interface, three status LEDs, common ground, and SELECT and START/ENTER buttons. The main SMPS power path, external high-current shunt integration, fuse/protection path, final PCB, enclosure, final firmware and built-in load are still under development or planned. A future version is intended to integrate a suitable internal electronic load with adequate heat dissipation and protection. The proposed system is therefore positioned as a low-cost, compact SMPS-focused bench diagnostic and testing instrument rather than a replacement for professional power analyzers.

Keywords: SMPS, power monitoring, voltage measurement, current measurement, INA226, ESP32, TFT LCD, electronic load, fault indication, embedded testing.

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Abbreviations and symbols

Abbreviation	Meaning
SMPS	Switched-Mode Power Supply
DC	Direct Current
ESP32	Espressif Systems 32-bit microcontroller family
INA226	16-bit I²C current, voltage and power monitor
TFT	Thin-Film Transistor display
LCD	Liquid Crystal Display
SPI	Serial Peripheral Interface
I²C	Inter-Integrated Circuit
GPIO	General Purpose Input/Output
PCB	Printed Circuit Board
TVS	Transient Voltage Suppressor
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
BOM	Bill of Materials
UI	User Interface
R&D	Research and Development
DUT	Device Under Test
WBS	Work Breakdown Structure
P	Power
V	Voltage
I	Current
R	Resistance
SDG	Sustainable Development Goal

CHAPTER 1:
INTRODUCTION

1.1 Background

Switched-mode power supplies are widely used because they can provide regulated DC power with relatively high efficiency and compact size. In an SMPS manufacturing and servicing environment, verifying the output of a supply is an important step before the unit is accepted, dispatched or subjected to further testing. Basic output parameters such as voltage and current provide an immediate indication of whether the supply is operating near its expected operating point.

Conventional laboratory testing can involve separate digital multimeters, current meters, electronic loads, oscilloscopes and power analyzers. Such equipment is useful and often necessary for detailed characterization, but a production or service technician may also benefit from a compact dedicated instrument that combines basic DC output measurement, local display and simple pass/warning/fault indication in one enclosure.

The proposed SMART SMPS MONITORING SYSTEM addresses this practical requirement. The system receives the DC output of an SMPS, measures voltage and current using a shunt-based INA226 measurement arrangement, calculates output power, displays the values on a TFT LCD and uses selectable profiles to determine a basic status. The design is intentionally focused on local operation; the final direction does not require a mobile application, Wi-Fi dashboard or cloud platform.

1.2 Problem Statement

The project addresses the following engineering problem: there is a need for a compact, low-cost and SMPS-focused instrument capable of performing basic DC output monitoring and providing an immediate local indication of whether measured output values are within the expected limits for a selected SMPS profile.

- Separate instruments may be required to observe voltage, current and power during routine SMPS checks.
- Technicians need a quick indication of abnormal output conditions during production, incoming inspection or troubleshooting.
- Generic meters normally do not provide a dedicated SMPS profile/threshold interface or integrated status indication.
- An internal load can make a tester more useful, but high-power load integration introduces thermal, wiring, connector, fuse and PCB design challenges.
- Broad voltage/current targets discussed during early planning cannot be directly claimed for the present prototype because the current INA226/protection architecture has lower limits.

1.3 Need of the Project

The need for the proposed system arises from the practical requirement for repeatable and convenient basic testing of DC-output SMPS units. A dedicated instrument can reduce the number of manual connections during routine checks and provide a consistent user interface. It can also support a profile-based testing approach, where the operator selects the expected output category before starting the measurement.

The project is particularly relevant to an SMPS manufacturing environment because the proposed instrument can be used as a first-level test device. It is not intended to replace a certified power analyzer or an oscilloscope-based qualification setup. Instead, it provides a focused combination of measurement, display and status indication for routine diagnostic work.

1.4 Objectives

General objective:

To design and develop a compact embedded SMART SMPS MONITORING SYSTEM for measuring DC output voltage, current and power and providing basic profile-based status indication.

Specific objectives:

- Design an ESP32-based controller for measurement processing and user interaction.
- Interface an INA226-based current/voltage/power measurement system using I²C communication.
- Measure the SMPS output voltage and load current through a properly rated external shunt arrangement.
- Calculate output power using the measured voltage and current.
- Display measurement values and test status on a 3.2-inch ILI9341 SPI TFT LCD.
- Provide green, yellow and red LED indication for normal, warning and fault states.
- Implement physical SELECT and START/ENTER controls for SMPS profile selection and test initiation.
- Develop a protection-oriented power path using suitable fuse, transient protection and switching components after component ratings are verified.
- Design the final PCB and enclosure with appropriate high-current routing and thermal management.
- Develop the architecture for a future integrated electronic load without claiming unverified high-power capability.

1.5 Scope

The scope covers the design of an embedded DC-output monitoring and basic testing instrument for SMPS applications. The system includes voltage/current measurement, power calculation, TFT display, physical controls and LED-based status indication. The final design may include an internal load/electronic load, but the high-power load stage is treated as a planned/final-stage subsystem rather than a completed prototype feature.

The present prototype scope is limited by the electrical ratings of the selected sensing and protection components. The INA226 used in the current architecture is a 36 V bus-voltage monitor. Therefore, direct 60 V measurement is outside the current prototype scope. Similarly, the available 5 A fuse, IRF4905 MOSFETs, SMBJ58A TVS components and 22 AWG wiring cannot be treated as sufficient for a 20 A/1200 W tester without redesign.

Future expansion may include higher-voltage sensing, higher-current sensing, programmable electronic load operation, data logging, calibration routines, automated test reports and advanced fault diagnostics. Such extensions require appropriate component selection and validation.

1.6 Expected Outcomes

- A working embedded prototype demonstrating ESP32 control, INA226 communication and TFT display integration.

- A local interface showing voltage, current, calculated power, selected profile and status.
- Three-level status indication through green, yellow and red LEDs.
- A structured profile/threshold mechanism for different SMPS output ratings.
- A documented high-current sensing and protection architecture suitable for PCB implementation.
- A final PCB/enclosure concept with a clearly defined path toward integrated load testing.
- An engineering report documenting limitations, safety considerations, testing methodology and future expansion.

CHAPTER 2:
LITERATURE REVIEW

2.1 Introduction

The literature survey was performed around four related areas: DC power-supply testing, programmable/electronic loads, SMPS condition monitoring and fault diagnosis, and embedded power measurement. The surveyed work shows that electronic loads are widely used to characterize power sources and that fault-diagnosis research can extract information from voltage/current or other sensed signals. However, many advanced systems focus on programmable loads, transient testing, waveform analysis or machine-learning-based diagnosis rather than a compact low-cost SMPS-focused instrument that combines basic measurement, local display and simple profile-based status indication.

2.2 Literature Review Table

Paper / Source	Method / System	Key Finding	Research Gap / Relevance
Chen, Qu & Ni, 2025	Programmable supply + electronic load	5–20 V supply and 0–10 A load with voltage/current feedback	Real-time measurement and load control can support repeatable product testing.
Chen, Qu & Ni, 2025	Programmable supply + electronic load; 5–20 V supply and 0–10 A load with voltage/current feedback.	Real-time measurement and load control can support repeatable product testing.	Oriented to USB/experimental testing; not a compact SMPS diagnostic instrument with simple local status classification.
Moras et al., 2024	Programmable electronic load with multiple operating modes for DC supply testing.	Demonstrates usefulness of programmable loading for simulating electrical conditions.	Does not focus on an integrated SMPS tester with dedicated profile selection and status LEDs.
Xu, Zhu & Wang, 2025	Dynamic wavelet packet transform + optimized SVM for SMPS fault diagnosis.	Sensed SMPS signals can support advanced fault classification.	Requires signal features, datasets and computational diagnosis beyond the current embedded threshold approach.
Tang et al., 2024/25	Flyback SMPS condition monitoring using input current and output voltage.	Combining electrical signals improves fault discrimination.	Advanced diagnosis is more complex than the compact first-level tester proposed here.
Liu & Cai, 2024	Wavelet features with PSO/CNN comparison for switching-supply faults.	Machine-learning models can classify multiple fault modes.	Data collection and model training are beyond the present project's scope.
Santoro et al., 2022	Cost-efficient DC active load for laboratory power-supply testing.	Highlights cost and thermal management as important in practical load design.	Primarily an active-load solution rather than an integrated SMPS monitoring/tester product.

Programmable electronic load study, 2022	Microcontroller-controlled programmable electronic load for equipment testing.	Demonstrates programmable current demand for experimental testing.	Focus is load emulation rather than simple production-oriented SMPS screening.
Novel DC electronic load topology, 2023	MPC-based DC electronic-load topology.	Advanced control can improve load behavior and flexibility.	Control complexity and power-stage requirements exceed the immediate prototype need.
IoT power-supply monitoring study, 2025	ESP32 + INA219 monitoring of 5/12/24 V supplies.	Demonstrates low-cost embedded monitoring of multiple DC supply levels.	Uses cloud/IoT concepts intentionally excluded from the final direction of this project.
Vlček, 2024/25	Programmable DC electronic load with CC/CR/CP, pulse and battery-test modes; MCU/LCD interface.	Shows practical integration of MCU, display, control and thermal design.	Focuses on a programmable load rather than a dedicated SMPS output tester.

2.3 Research Gap

The surveyed work indicates that significant research exists in programmable electronic loads and advanced SMPS fault diagnosis. However, there is a practical design gap between high-end laboratory instruments and simple single-purpose meters. The proposed project addresses this gap by combining a dedicated SMPS test workflow with embedded voltage/current/power measurement, a local TFT interface, profile-based thresholds and three-level status indication.

The project emphasizes compactness and direct local operation instead of cloud monitoring.

- The first-level health/status decision is based on measured output parameters and selected SMPS limits rather than computationally intensive fault classification.
- The design considers the practical integration of measurement, protection, user interface and enclosure.
- The future built-in load concept is considered together with its thermal and high-current design constraints.
- The system is intended as a low-cost complementary instrument rather than a replacement for professional power analyzers.

2.4 Novelty / Innovation

- SMPS-specific workflow rather than a generic voltmeter/ammeter.
- Profile-based test selection using physical buttons.
- Combined display of voltage, current, power and status on one TFT interface.
- Immediate visual classification using green/yellow/red LEDs.
- Industry-oriented design considering shunt selection, protection, PCB current capacity, connector rating and thermal management.
- Architecture that can later incorporate a programmable internal load.
- Explicit engineering limitation management so unsupported voltage/current ratings are not presented as prototype capability.

CHAPTER 3: PROPOSED METHODOLOGY

3.1 Proposed System

The proposed system consists of an input connection from the DC output of an SMPS under test, a protected measurement path, a shunt-based current sensing stage, the INA226 measurement device, an ESP32 controller, a TFT display, user push buttons and LED status indicators. A regulated low-voltage supply path using the LM2596 module may be used where required by the final power architecture.

The current prototype has already established the digital control/display section: ESP32-to-INA226 I²C communication, ESP32-to-TFT SPI communication, common ground, three status LEDs, SELECT and START/ENTER buttons. The high-current power path and final measurement integration are still under development.

3.2 Working Principle

1. Connect the DC output of the SMPS under test to the tester input through the final protected input path.
2. Select the appropriate SMPS profile using the SELECT button.
3. Press START/ENTER to begin the test sequence.
4. The INA226 measures the bus voltage and the voltage drop across the current shunt.
5. The ESP32 reads the measurement data over I²C.
6. The firmware calculates output power as P = V × I.
7. The measured values are compared with the limits stored for the selected SMPS profile.
8. The TFT displays voltage, current, power, profile and status.
9. The green, yellow or red LED is activated according to the classification.
10. A future safe buzzer driver can provide an audible indication for warning/fault conditions.
11. The result can be reset or another profile can be selected for the next test.

3.3 System Architecture / Block Diagram

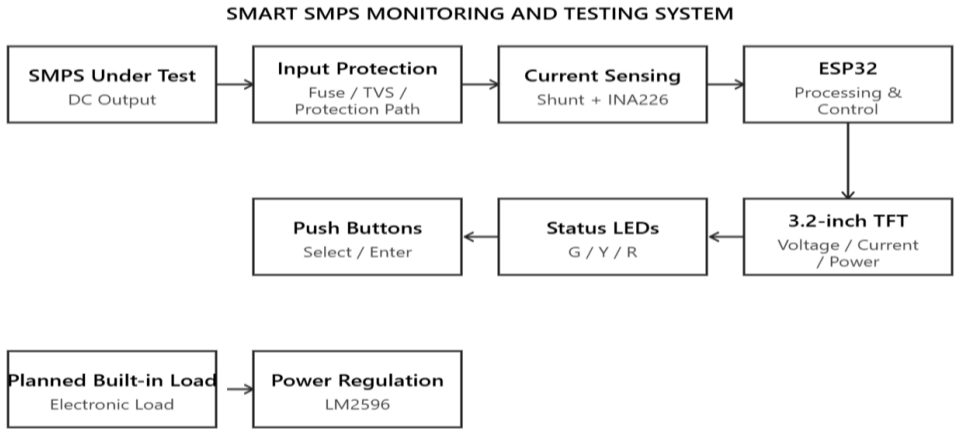


Figure 2.1 – Proposed architecture of the SMART SMPS MONITORING SYSTEM

The architecture separates the high-current power path from the low-voltage control and display electronics. This separation is important because the measurement shunt, PCB copper, terminals, fuse, switching devices and load stage must be rated for the actual current, while the ESP32 and TFT operate at low logic voltage.

3.4 Flow Chart

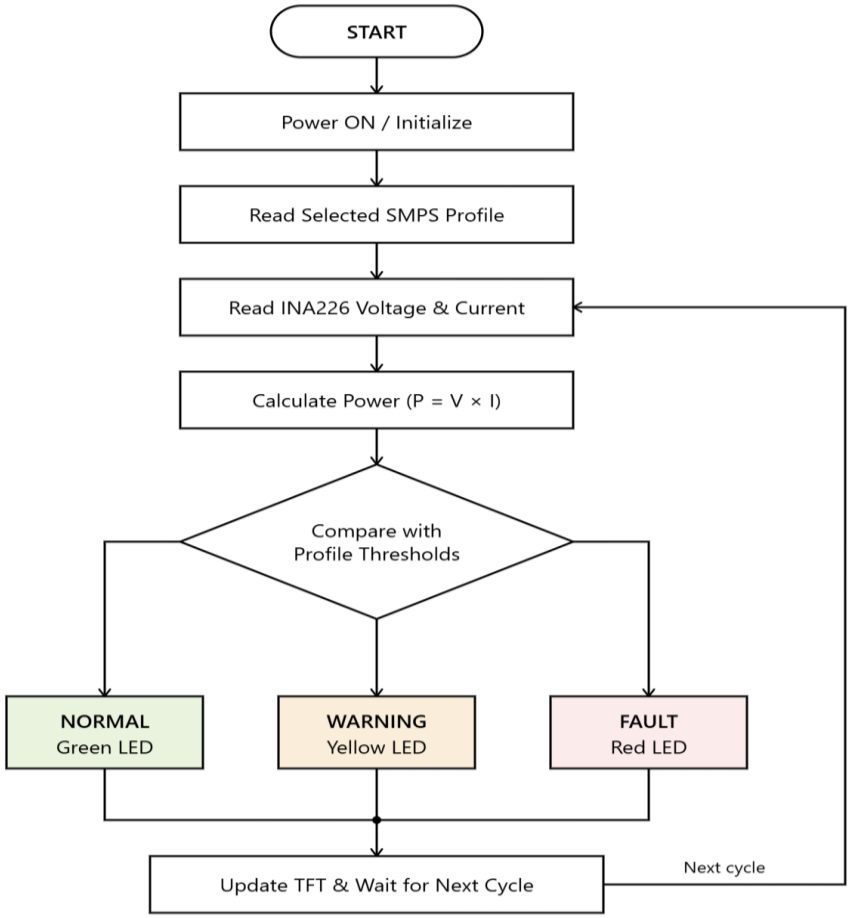


Figure 2.2 – Firmware flowchart for measurement and status classification

3.5 Methodology

3.5.1 Hardware Design Procedure

- Define the required SMPS test profiles and their nominal voltage/current values.
- Select the measurement architecture and confirm the voltage/current limits of the sensing device.
- Select an external shunt with a known resistance and suitable power rating.
- Design the protected input path and confirm fuse, TVS, MOSFET, terminal and wire ratings.
- Integrate the INA226 into the high-side measurement path.
- Connect the INA226 to the ESP32 using I²C.
- Connect the TFT to the ESP32 using SPI.
- Connect LEDs and physical push buttons to the allocated GPIO pins.

- Develop the PCB layout with high-current and logic sections separated as appropriate.
- Provide suitable mounting and cooling for the planned internal load.

3.5.2 Software Procedure

- Initialize GPIO, I²C, SPI and TFT display.
- Initialize the INA226 and apply the correct shunt calibration after the external shunt specification is confirmed.
- Read voltage and current at regular intervals.
- Calculate power using the measured values.
- Read the selected SMPS profile from the button interface.
- Compare measurements with profile thresholds.
- Update the TFT display and LED status.
- Provide fault/warning text and optional buzzer control through a safe transistor/MOSFET driver in the final version.
- Repeat the measurement cycle until the user stops or changes the test.

3.5.3 Experimental Procedure

- Begin with a low-voltage current-limited test source rather than an unverified high-power SMPS.
- Verify the digital communication between ESP32 and INA226 before connecting the main power path.
- Verify TFT display and LED indication independently.
- Confirm the external shunt resistance and power rating from its marking/datasheet.
- Connect the protected measurement path using properly rated conductors and terminals.
- Compare measured voltage/current with a calibrated or trusted reference meter.
- Test multiple known operating points within the verified prototype range.
- Record measurement error, display response and status classification.
- Increase load only within the verified current, voltage and thermal limits.
- Document any deviations and update calibration constants before final testing.

3.6 Components / Materials

Component	Specification / Role	Qty.	Status
ESP32 DevKit V1 / ESP-WROOM-32	32-bit MCU; I²C, SPI, GPIO	1	Implemented
INA226 module	16-bit high-side current/voltage/power monitor; 36 V bus limit	2	Digital interface implemented; power path pending
External high-current shunt	Low-value precision current-sense resistor; exact rating TBD	1	Purchased / pending integration
ILI9341 TFT LCD	3.2 in, 320×240, SPI	1	Implemented
LM2596 module	Adjustable buck converter; nominal module use for low-voltage supply	1	Available; final role pending

Active buzzer	5 V	1	Available; driver pending
Green LED	Normal indication	1	Implemented
Yellow LED	Warning indication	1	Implemented
Red LED	Fault indication	1	Implemented
1 kΩ resistors	LED current limiting	3+	Implemented
6×6 mm tactile push buttons	SELECT / START-ENTER	2+	Two implemented
5 A slow-blow fuse + holder	Input protection for appropriate low-current path only	1	Available; final rating TBD
SMBJ58A TVS	Transient protection component	As required	Available; not a 60 V continuous-operation solution
IRF4905 P-MOSFET	High-side switching/protection candidate	As required	Available; rating must be verified for final use
KF301-2P terminal	Screw terminal	As required	Available
22 AWG silicone wire	Low/medium-current wiring	As required	Available; not for 20 A path
ABS enclosure	Mechanical enclosure	1	Planned
M3 hardware	PCB/enclosure mounting	As required	Available
FR-4 PCB	Final circuit board	1	Planned

3.7 Measurement Methodology and Calculations

The current measurement uses a shunt resistor placed in series with the load path. The voltage drop across the shunt is measured by the INA226. The current is derived from the shunt voltage and the configured shunt resistance. The INA226 also measures bus voltage and internally supports power computation after calibration.

For a basic independent verification, the current can be expressed as:

$I = V_{shunt} / R_{shunt}$

where V_{shunt} is the voltage measured across the external shunt and R_{shunt} is the verified shunt resistance.

The output power is calculated by:

$P = V \times I$

For example, a display example may show 24.00 V, 2.35 A and 56.40 W; these values are only a user-interface example and are not claimed as measured project results.

3.8 SMPS Profile and Threshold Logic

The current rating of an SMPS cannot be determined simply by observing the instantaneous current drawn by a connected load. Therefore, the tester should not infer the nameplate current rating automatically from one measurement. Instead, the user selects a predefined profile or enters a known expected rating in the final interface.

A practical threshold scheme can use a voltage tolerance band and a current/power operating band. Exact threshold percentages should be selected according to the SMPS manufacturer's specification rather than

assumed universally. The synopsis therefore defines the method, while final numeric thresholds are to be configured from actual product specifications

Profile	Nominal Rating	Expected V	Expected I	Engineering Note
Profile A	5 V / 2 A	5 V	2 A	Profile values to be validated before final testing
Profile B	12 V / 5 A	12 V	5 A	Profile values to be validated before final testing
Profile C	24 V / 10 A	24 V	10 A	Nominal example only; current prototype does not claim universal 10 A operation

3.9 Fault Indication Strategy

The proposed first-level status classifier is intentionally simple. A normal state is assigned when measured parameters are within the selected profile limits. A warning state can be assigned when values approach a defined limit or when a non-critical deviation is observed. A fault state can be assigned when voltage/current exceeds a critical threshold, the output is absent when it should be present, or another defined abnormal condition is detected.

Condition	Indication	Purpose
Measured output within validated limits	Green LED / NORMAL	Pass or normal operating state
Measurement near warning boundary	Yellow LED / WARNING	Operator attention required
Critical over/under-voltage, overcurrent or missing output	Red LED / FAULT	Stop or reject condition
Sensor communication/calibration error	Red LED / SENSOR FAULT	Measurement cannot be trusted

3.10 PCB, Protection and Thermal Design

The final PCB must separate low-voltage digital circuitry from high-current copper paths as far as practical. The external shunt should be placed so that current flows through the intended high-current path and the INA226 sense connections use short, controlled connections. High-current connectors and terminals must be rated above the intended operating current, and copper width/thickness must be selected based on allowable temperature rise.

The available SMBJ58A TVS and IRF4905 components should not be used as justification for a 60 V continuous-input product. The current INA226 has a 36 V bus limit. Similarly, 22 AWG wiring is not appropriate for a 20 A main path. The final high-power load must use power-rated components, heat sinking, airflow/ventilation where necessary and an enclosure that can safely dissipate heat.

The DS18B20 temperature sensor was originally considered but removed from the current design to simplify the prototype. Temperature monitoring may be reintroduced in a future version, especially for the internal load stage.

CHAPTER 4:
PROJECT PLANNING

4.1 Work Breakdown Structure

WBS Item	Activities
1. Requirement analysis	1.1 Industrial requirements; 1.2 SMPS profiles; 1.3 safety limits
2. Hardware design	2.1 Controller; 2.2 sensing; 2.3 display; 2.4 controls; 2.5 protection
3. Prototype integration	3.1 wiring; 3.2 I²C; 3.3 SPI; 3.4 LEDs; 3.5 buttons
4. Measurement path	4.1 shunt verification; 4.2 INA226 calibration; 4.3 protected power path
5. Firmware	5.1 drivers; 5.2 calculations; 5.3 profile logic; 5.4 UI; 5.5 fault logic
6. PCB and enclosure	6.1 schematic; 6.2 layout; 6.3 fabrication; 6.4 mounting
7. Load subsystem	7.1 load selection; 7.2 thermal design; 7.3 integration
8. Testing	8.1 reference comparison; 8.2 profile testing; 8.3 fault testing; 8.4 documentation
9. Final documentation	9.1 synopsis; 9.2 report; 9.3 presentation; 9.4 demonstration

4.2 Gantt Chart

Activity	Planned Period	Status
Requirement & literature survey	June	Completed
Component verification	July	Completed
ESP32 + INA226 + TFT integration	July	Completed
LED + button interface	August	Completed
Power path + shunt integration	August	Ongoing
Protection design	September	Ongoing
Firmware	September	Completed
PCB schematic/layout	September	Planned
Enclosure + internal load design	October	Planned
Calibration and validation	October	Planned
Final testing and documentation	October	Planned

4.3 Roles and Responsibilities

Member / Role	Responsibility
Mariyam Bag	Project coordination, system integration, documentation and testing support
Dhruv Bari	Hardware interfacing, measurement path and PCB support
Vikas Kanojiya	ESP32 firmware, TFT UI and status logic
Mangal Dwivedi	Testing, BOM/procurement support, enclosure and

	documentation
Project Guide / Industrial Support	Technical review, PCB guidance and validation support

4.4 Risk Assessment

Risk	Impact	Mitigation
INA226 over-voltage exposure	High	Restrict prototype bus voltage to verified limit; redesign sensing for higher voltage
Incorrect shunt rating	High	Verify resistance and power rating before calibration and high-current tests
Excessive shunt heating	High	Use low-value power-rated shunt and calculate I^2R dissipation
Insufficient wire/PCB current capacity	High	Use rated wire, terminals, copper area and thermal analysis
Fuse incorrectly sized	High	Select fuse based on actual current and fault-clearing requirement
MOSFET/TVS voltage margin insufficient	High	Use components with adequate voltage margin for the final target
TFT/ESP32 power instability	Medium	Use regulated supply, common grounding and decoupling
Button debounce errors	Low	Implement software debounce
False warning/fault indication	Medium	Validate thresholds using known reference supplies
Internal load overheating	High	Power-rated load, heatsink, airflow and thermal cut-off
Accidental high-power connection	High	Use labelled terminals, fuse/protection and staged commissioning
Buzzer GPIO overload	Medium	Use transistor/MOSFET driver and verify buzzer current

CHAPTER 5: RESOURCES & FEASIBILITY

5.1 Technical Feasibility

The project is technically feasible because the core measurement and control functions can be implemented using commercially available embedded modules. ESP32 provides I²C, SPI and GPIO interfaces required for the INA226, TFT and controls. The INA226 is designed for current, voltage and power monitoring and provides a digital I²C/SMBus-compatible interface. The main technical challenge is not the digital interface but the safe design of the high-current measurement and load path.

The technical feasibility of higher-voltage operation is conditional. The current INA226-based architecture cannot be used directly at 60 V because the device's bus-voltage limit is 36 V. Therefore, a future higher-voltage version would require a different sensing architecture or appropriate voltage scaling and protection.

5.2 Economic Feasibility

The proposed system uses low-cost embedded modules and standard components. The largest cost drivers are expected to be the final PCB, external shunt, high-power load components, thermal hardware and enclosure. Compared with professional power analyzers and programmable electronic-load systems, the proposed instrument is intended to provide a lower-cost solution for basic SMPS screening and diagnostic tasks.

5.3 Operational Feasibility

The system is intended for operation by a technician with basic electrical test knowledge. A simple sequence of selecting an SMPS profile, connecting the output, starting the test and observing the TFT/LED status reduces unnecessary interaction. However, users must still follow safe test procedures because the instrument handles electrical energy and may be connected to high-current sources.

5.4 Budget Estimation

Item	Quantity	Indicative Cost	Note
ESP32 DevKit V1	1	₹400–₹700	Indicative
INA226 modules	2	₹500–₹1,000	Indicative
3.2-inch ILI9341 TFT	1	₹350–₹700	Indicative
External shunt	1	₹150–₹500	Depends on rating
LM2596 module	1	₹60–₹150	Indicative
LEDs/resistors/buttons	Set	₹100–₹250	Indicative
Fuse, holder, TVS, MOSFETs	Set	₹250–₹600	Indicative
Terminals/wire/headers	Set	₹250–₹500	Indicative
ABS enclosure + hardware	Set	₹250–₹600	Indicative
FR-4 PCB fabrication	1	₹500–₹1,500	Depends on size/vendor
Internal load + heatsink/fan	Set	₹800–₹2,500+	Highly dependent on target power
Miscellaneous	—	₹300–₹700	Indicative

Indicative prototype budget: approximately ₹3,900 to ₹9,700+, excluding specialized laboratory instruments and depending strongly on the final load stage and PCB fabrication. These are planning estimates, not purchase invoices, and should be replaced by actual procurement costs in the final report.

5.5 Bill of Materials (BOM)

Component	Qty.	Purpose	Status
ESP32 DevKit V1 / ESP-WROOM-32	1	Control processor	Available
INA226 CJMCU-226 type	2	Voltage/current/power sensing	Available
External high-current shunt	1	Main current sensing	Available; rating TBD
ILI9341 3.2-inch TFT	1	Local display	Available
LM2596 adjustable buck	1	Low-voltage regulation	Available
Active buzzer 5 V	1	Audible warning	Available; driver needed
Green/yellow/red LED	1 each	Status indication	Available
1 kΩ resistors	3+	LED limiting	Available
Tactile push buttons	2+	User input	Available
5 A slow-blow fuse + holder	1	Protection	Available
SMBJ58A TVS	As required	Transient protection	Available; application-dependent
IRF4905	As required	Switching/protection candidate	Available; rating-dependent
KF301-2P terminals	As required	Power connection	Available
22 AWG silicone wire	As required	Wiring	Available; not 20 A path
ABS enclosure	1	Mechanical protection	Planned
M3 hardware	Set	Mounting	Available
FR-4 PCB	1	Final assembly	Planned
High-power load stage	1	Built-in testing load	Planned

CHAPTER 6: EXPECTED RESULTS

This synopsis intentionally presents expected results and validation criteria rather than fabricated experimental results. Actual numerical accuracy, thermal performance and load capability will be reported only after the complete measurement path has been integrated and tested.

6.1 Expected Prototype

- The prototype is expected to read DC output voltage and current through the INA226 measurement system.
- The ESP32 is expected to calculate output power and update the TFT display.
- The user is expected to select an SMPS profile using the physical controls.
- The green/yellow/red LEDs are expected to provide immediate status indication.
- The final enclosure is expected to provide a compact and safe physical arrangement of the electronics.
- A future final version is expected to incorporate a suitable internal load after power and thermal design validation.

6.2 Expected Accuracy and Performance

The expected measurement performance will be determined experimentally against reference instruments. The INA226 itself is a precision current/voltage/power monitor, but system-level accuracy depends on the selected shunt, PCB layout, wiring resistance, calibration, connector losses and reference instrument accuracy. Therefore, no fixed percentage accuracy is claimed at the synopsis stage.

Expected validation parameters include:

- Voltage measurement error relative to a reference digital multimeter.
- Current measurement error relative to a calibrated reference meter or load measurement.
- Power calculation consistency using $P = V \times I$.
- Display update stability and responsiveness.
- Correct profile selection and threshold classification.
- Protection behavior under controlled abnormal conditions.
- Thermal stability of the shunt, load stage and power-path components.

6.3 Expected Testing Procedure

Test Case	Condition	Expected Verification
No-load / minimum load	Within verified voltage range	Check output voltage and status
Low current	Known current below nominal	Compare current and power with reference
Mid-range current	Controlled load	Check measurement linearity and status
Near profile limit	Controlled load	Verify warning threshold
Abnormal voltage	Controlled safe test	Verify fault indication
Overcurrent simulation	Controlled and protected	Verify fault response; do not exceed component ratings

Sensor disconnect	Safe controlled condition	Verify sensor-fault handling
Repeated cycles	Multiple start/stop operations	Check UI and firmware stability

6.4 Industrial Relevance

The system can serve as a first-level SMPS output verification instrument in production, incoming inspection and R&D environments. It can reduce the need to manually observe several instruments for routine checks and can provide a standardized operator interface. The instrument is not intended to replace detailed compliance testing, ripple/noise analysis, transient response testing or certified power analysis.

CHAPTER 7:
SUSTAINABILITY & IMPACT

7.1 Industrial Relevance

The strongest impact of the project is industrial. SMPS manufacturers and service teams routinely need to verify output parameters. A compact tester can support repetitive checks and troubleshooting while providing a consistent test procedure. The architecture also creates a platform for future integration of an internal programmable load and automated test records.

7.2 Societal Impact

The project provides an educational and practical example of embedded instrumentation, power electronics measurement and product-oriented engineering. By focusing on repair, verification and diagnostics, the instrument can potentially reduce unnecessary replacement of otherwise serviceable power supplies. The project also develops engineering skills in safe handling of electrical energy and hardware-software integration.

7.3 Environmental Impact

A diagnostic tester can support maintenance and repair, potentially extending the useful life of SMPS units and reducing electronic waste. The final product should use a durable enclosure and serviceable PCB where practical. Thermal losses from a built-in load must be considered because inefficient dissipation converts electrical energy into heat.

7.4 SDG Mapping

SDG	Project Relevance
SDG 7 – Affordable and Clean Energy	Supports efficient use and monitoring of electrical power equipment.
SDG 9 – Industry, Innovation and Infrastructure	Directly supports industrial instrumentation and embedded-system innovation.
SDG 12 – Responsible Consumption and Production	Testing and repair can support longer equipment life and reduced waste.
SDG 13 – Climate Action	Indirect contribution through improved equipment maintenance and reduced avoidable replacement.

7.5 Future Scope

- Redesign the input sensing stage for higher-voltage SMPS outputs using appropriate isolation/scaling and protection.
- Develop a higher-current measurement path using a verified low-value shunt and suitable amplifier/sensor architecture.
- Integrate a programmable electronic load with constant-current, constant-resistance or constant-power modes.
- Add temperature sensing for the load and power-stage components.
- Implement thermal protection and automatic load shutdown.
- Add data logging to local storage or a PC interface if required by the industrial application.
- Add automatic calibration and service/calibration records.
- Expand fault classification using waveform features or machine-learning methods after sufficient test data are collected.

- Develop a production-line version with standardized connectors and automated pass/fail reporting.
- Design a higher-power PCB and enclosure with appropriate cooling, creepage/clearance and protection.

CHAPTER 8:
CONCLUSION

SMART SMPS MONITORING SYSTEM is proposed as a compact, industry-oriented embedded instrument for basic DC-output monitoring and testing of switched-mode power supplies. The project combines an ESP32 controller, INA226-based voltage/current/power measurement, a TFT display, physical controls and three-level status indication in a single system concept.

The project addresses the practical need for a dedicated first-level SMPS testing instrument that is more integrated than separate handheld measurements while remaining simpler and less costly than professional power-analysis equipment. The current prototype has already established the core digital interfaces, including ESP32-to-INA226 I²C communication, TFT SPI communication, status LEDs and two push buttons.

The remaining engineering work focuses on the high-current power path, external shunt integration, protection, calibration, firmware completion, PCB design, enclosure integration and the planned internal load. The project deliberately avoids unsupported claims: the current INA226 architecture has a 36 V bus-voltage limit, and the available protection/wiring components do not establish safe 60 V/20 A operation. Consequently, higher-voltage and higher-current capability is identified as future work requiring a redesigned architecture.

The expected contribution is a practical, low-cost SMPS-focused diagnostic and testing platform that can be further developed for production testing, incoming inspection, laboratory diagnostics and troubleshooting. The final evaluation will be based on measured accuracy, repeatability, protection behavior, thermal performance and successful operation within the validated design limits.

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